

Mechatronics Tutorial Sheet 4: Frequency response and Bode diagrams

1. A sinusoidal input is applied to a system with transfer function

$$H(s) = \frac{5}{(s+1)(s+10)}.$$

At which frequency does the output lag behind the input by exactly 90° ? What is the gain of the system at this frequency?

$$\text{Let } s = j\omega. \quad H(s) = \frac{5}{(j\omega+1)(j\omega+10)} = \frac{5}{(-\omega^2+10)+11j\omega}$$

When the case pole is purely imaginary, $\phi = -90^\circ$

$$\therefore -\omega^2 + 10 = 0 \quad \therefore \omega = \sqrt{10}$$

$$\therefore |H(s)| = \left| \frac{5}{11\sqrt{10}j} \right| = \frac{5}{11\sqrt{10}} = \frac{\sqrt{10}}{22} = 0.1437$$

$$= 20 \log_{10} \left(\frac{5}{11\sqrt{10}} \right) (\text{dB}) = -16.8425 (\text{dB})$$

2. Draw the Bode diagram for each of the following systems:

i. $H(s) = \frac{(s+1)}{s(s+100)}$

ii. $H(s) = \frac{s(-s+10)}{(s+1)^2}$

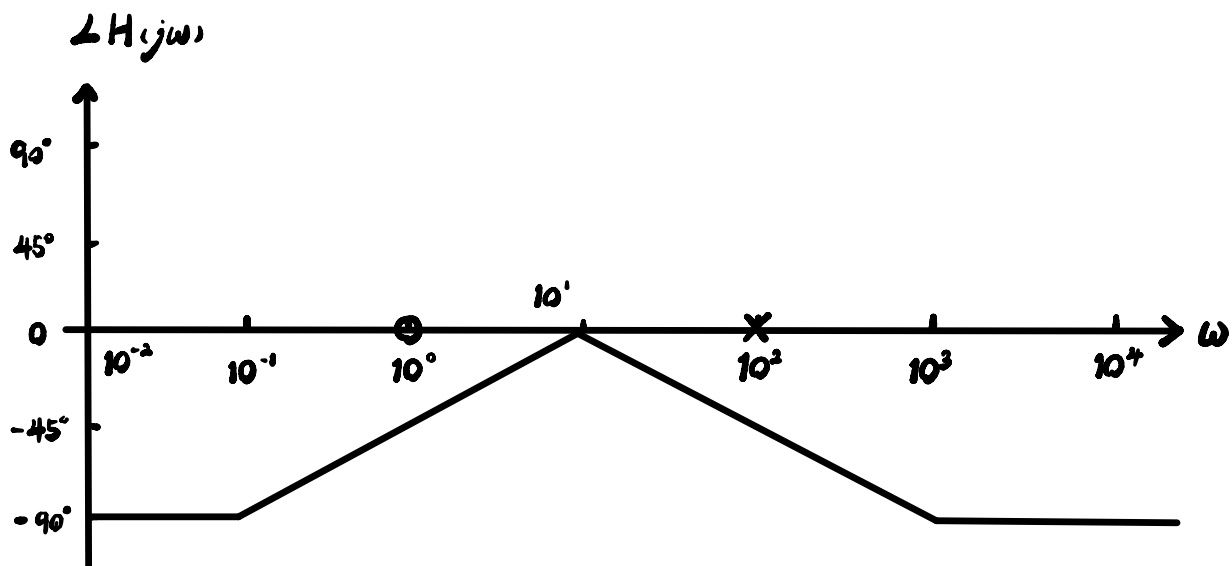
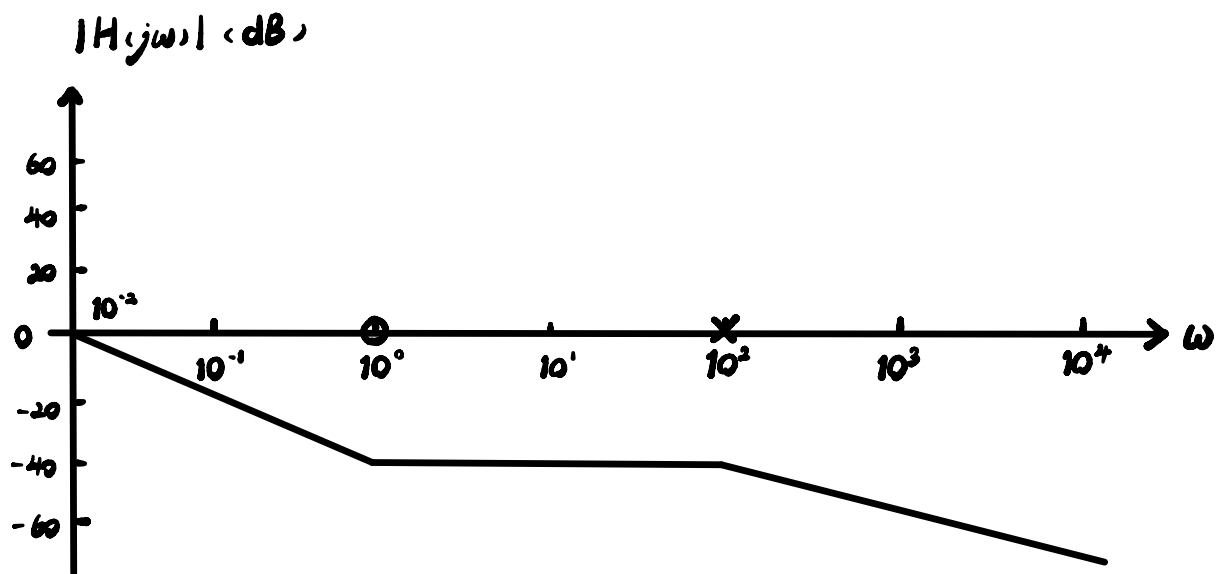
iii. $H(s) = \frac{50(s-10)}{(s+10)(s+100)}$

iv. $H(s) = \frac{1}{s(1+csT+s^2T^2)}$, where $c = 0.01$ and $T = 0.1$

i. zero: $s = -1$, poles: $s = 0$ and -100

for $s = j\omega$, for $\omega = 10^{-2}$, $H(s) = \frac{0.01j+1}{0.01j(0.01j+100)} \approx \frac{1}{0.01j(100)}$

$\therefore |H(s)| \approx 1 = 0 \text{ dB}$, $\angle H(s) = -90^\circ$



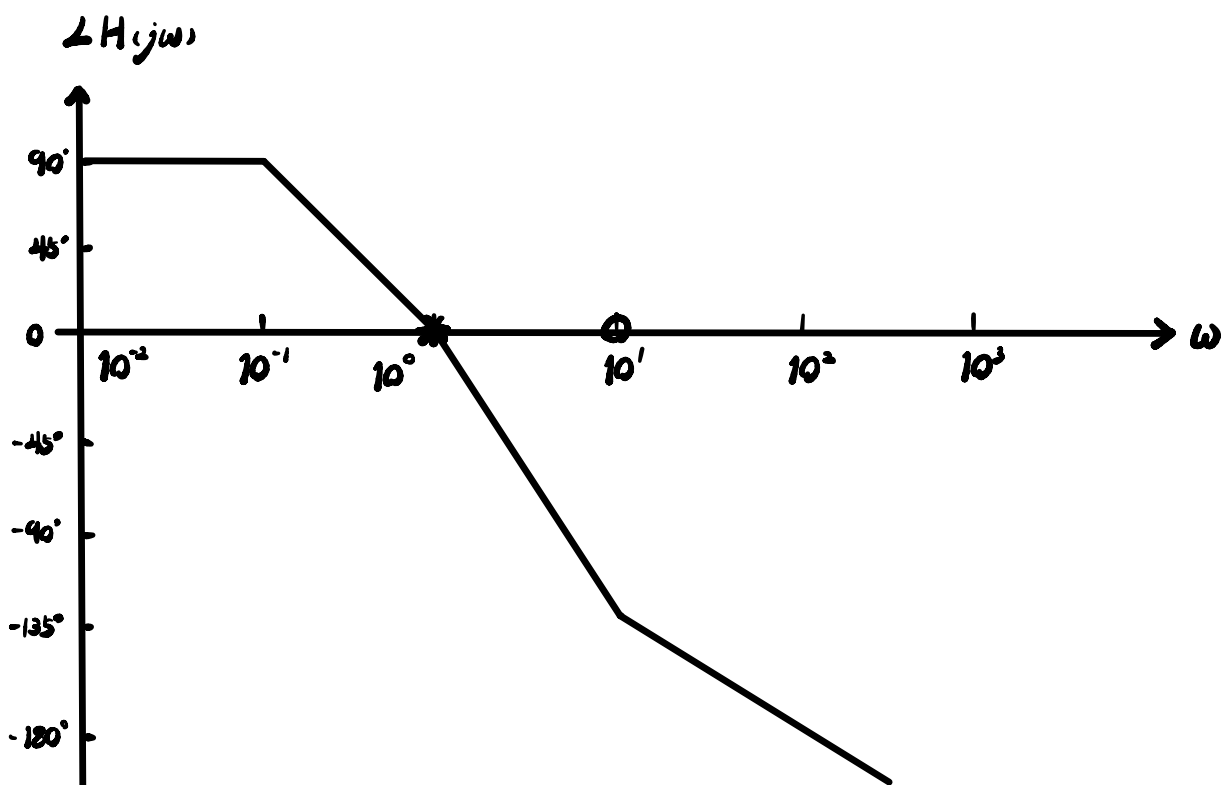
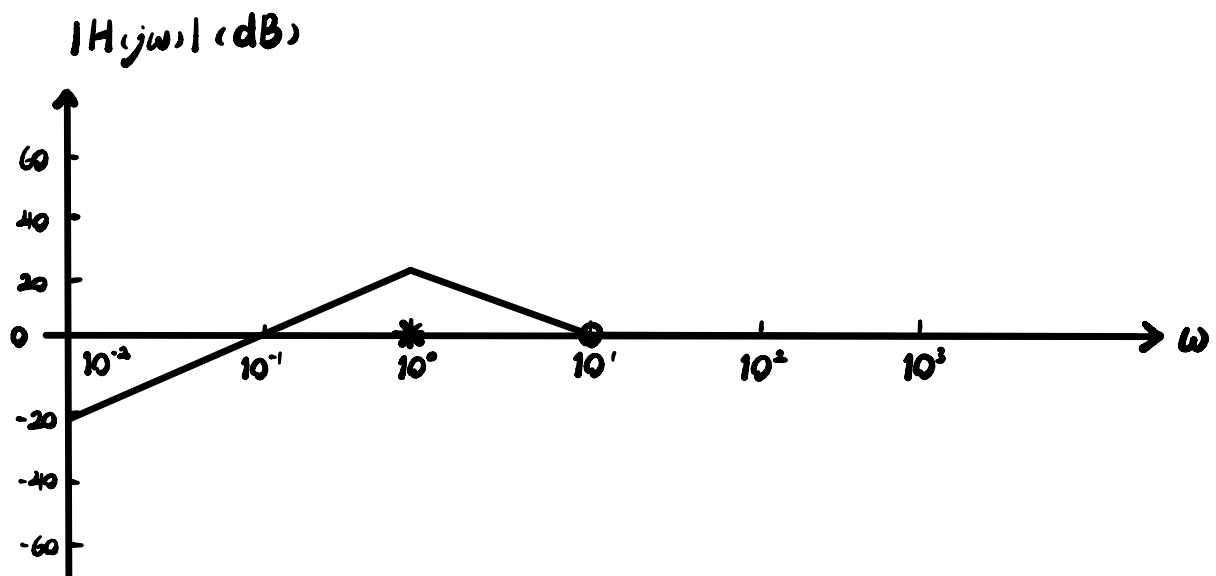
ii. zeros : $s = 0$ and $s = 10$

poles : $s = -1$ (double)

for $s = j\omega$, at $\omega = 10^{-2} = 0.01$.

$$H(j\omega) = \frac{0.01j(-0.01j+10)}{(0.01j+1)^2} \approx \frac{0.01j(10)}{1^2} = 0.1j$$

$$\therefore |H(j\omega)| = 0.1 = -20 \text{ dB}$$



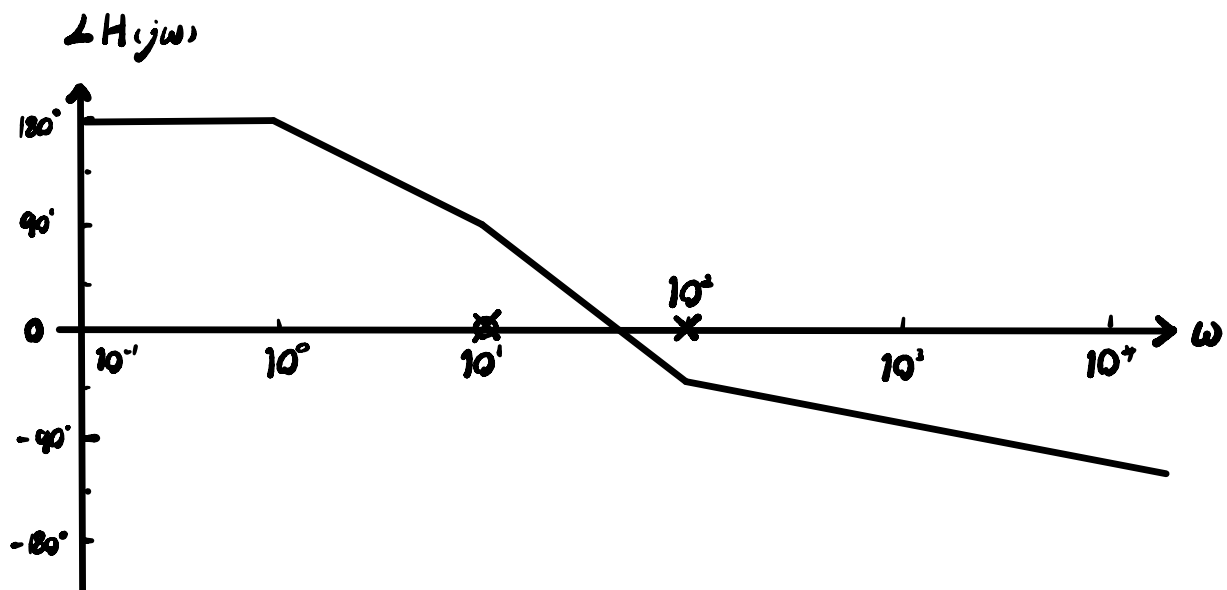
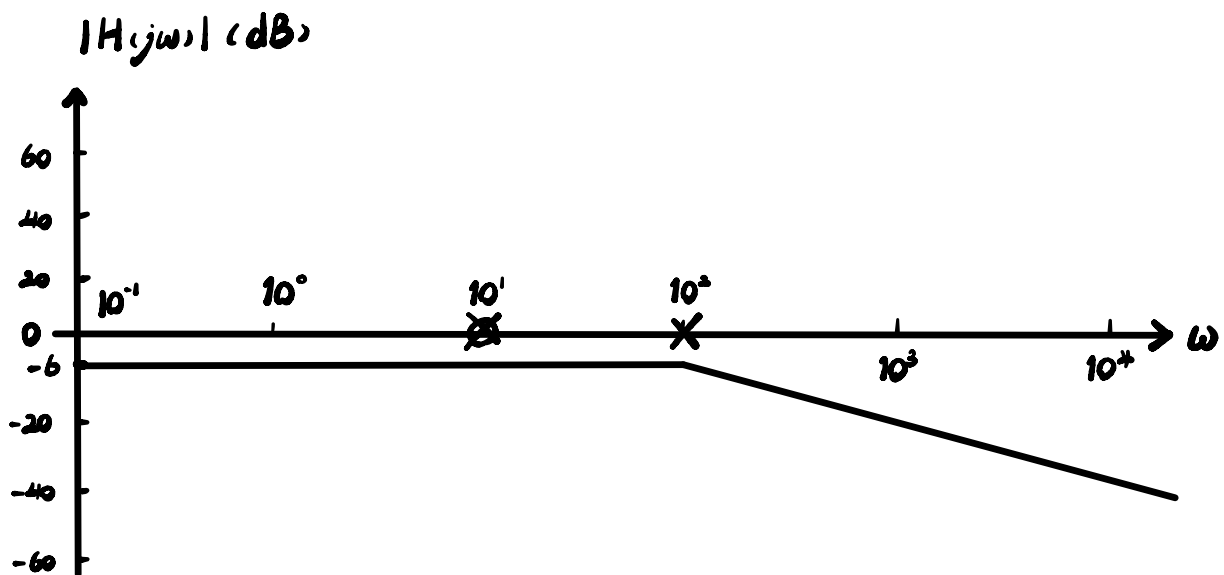
iii. zeros : $s = 10$

poles : $s = -10$ and $s = -100$

for $s = j\omega$, at $\omega = 10^{-1} = 0.1$,

$$H(0.1j) = \frac{50(0.1j - 10)}{(0.1j + 10)(0.1j + 100)} \approx \frac{50 \cdot (-10)}{10 \cdot 100} = -0.5$$

$$\therefore |H(0.1j)| = 0.5 = -6 \text{ dB}$$



$$\text{iv. } H(s) = \frac{1}{s(1 + 0.001s + 0.01s^2)}$$

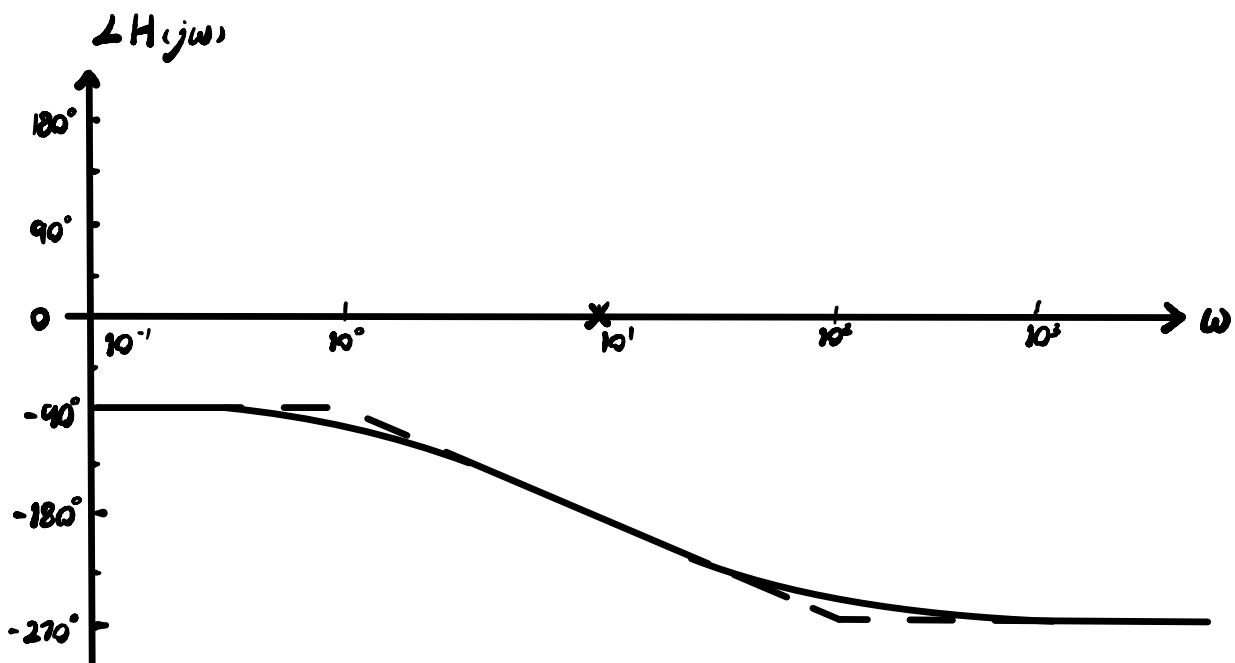
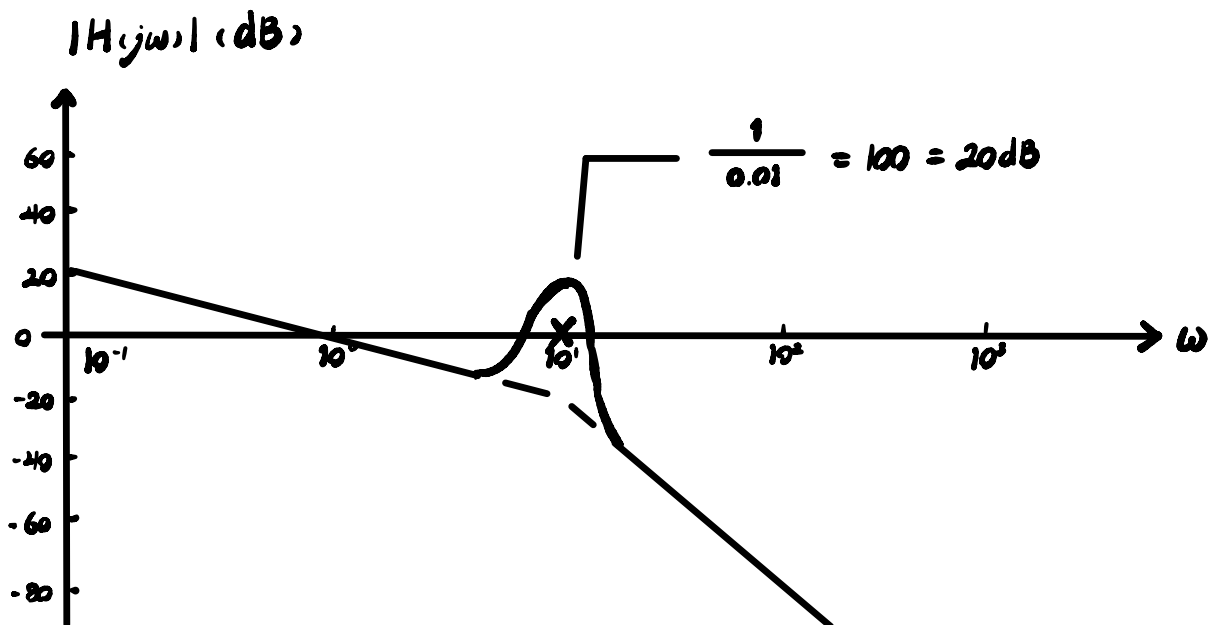
zeros: none

$$\text{poles: } s = 0 \text{ and } s = \frac{-0.001 \pm \sqrt{-0.039999}}{2} = -0.0005 \pm \frac{1}{2}\sqrt{0.039999}i$$

$$\therefore \omega_b = 0 \text{ and } \frac{1}{T} = 10$$

$$\text{for } s = j\omega, \omega = 0.1, H(0.1j) \approx \frac{1}{0.1j} = -10j$$

$$\therefore |H(j\omega)| = 10 = 20\text{dB}$$



3. For each of the Bode diagrams shown in Figure 3 (see overleaf), find the corresponding system transfer function.

Figure 3.A.

$$\omega_b = 10^1 \text{ and } 5 \times 10^1 \text{ and } 10^2$$

$$\therefore H(s) = K \frac{1 - 2\xi_z T_z s + T_z^2 s^2}{(1 + 2\xi_p T_p s + T_p^2 s^2) \left(\frac{s}{100} + 1 \right)}$$

$$\text{where } T_p = 0.1 \text{ and } T_z = 0.02$$

$$\text{At } \omega = 10^0 = 1, H(j) \approx K = 34 \text{ dB}$$

$$\therefore K = 10^{\frac{34}{20}} = 50.11$$

$$\text{At } \omega_b = 10^1, |H(j)| \approx \frac{K}{2\xi_p} = 60 \text{ dB} = 1000 \quad \therefore \xi_p = 0.025$$

$$\text{Similarly, } \xi_z = 0.025$$

break frequency :

$$\omega = 10' \rightarrow 2 \text{ poles at } s = -10$$

$$\omega = 5 \times 10' \rightarrow 2 \text{ zeros at } s = 50$$

$$\omega = 10^2 \rightarrow 1 \text{ pole at } s = 100$$

Therefore,

$$H(s) = \frac{K (1 - 2\xi_z s T_z + s^2 T_z^2)}{(1 + 2\xi_p s T_p + s^2 T_p^2)(s - 100)}$$

$$\text{where } T_p = \frac{1}{10}, T_z = \frac{1}{50}$$

$$\therefore \text{At } s = j, H(s) \approx K$$

$$\therefore |H(j)| = K = 10^{\frac{35}{10}} = 56.23$$

$$\therefore |H(10j)| \approx \left| 56.23 \frac{1}{1 + 2j\xi_p} \right| = 10^{\frac{62}{10}} = 1000$$

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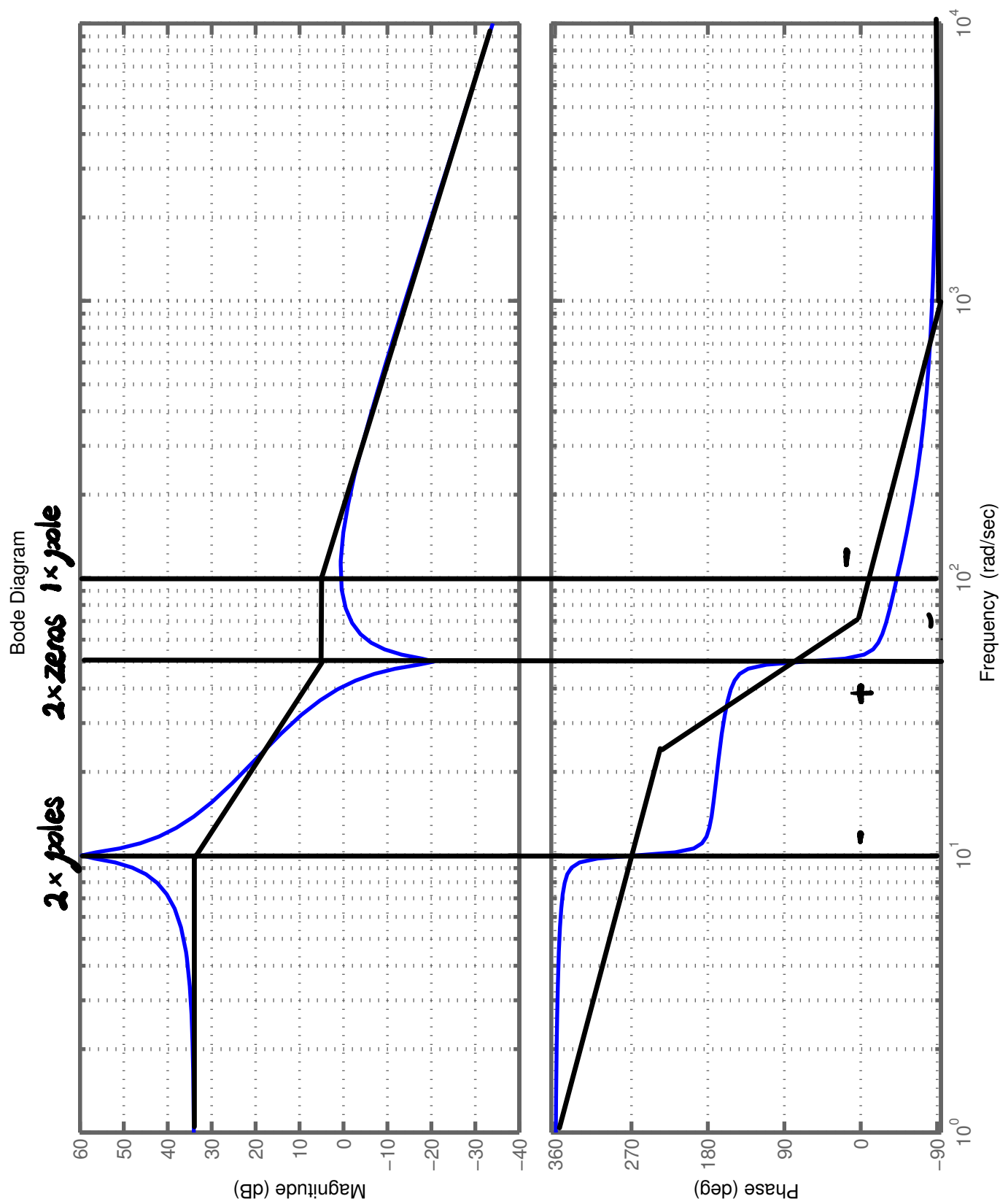


Figure 3.A

Figure 3.B.

$\omega_b = 0, 0.1$ and 10

$$\therefore H(s) = K \frac{(10s + 1)^2}{s \left(\frac{s}{10} + 1 \right)}$$

for $\omega = 10^{-2} = 0.01$.

$$H(0.01j) = \frac{K}{0.01} = 46 \text{ dB}$$

$$\therefore K = 0.01 \cdot 10^{\frac{46}{20}} = 1.995$$

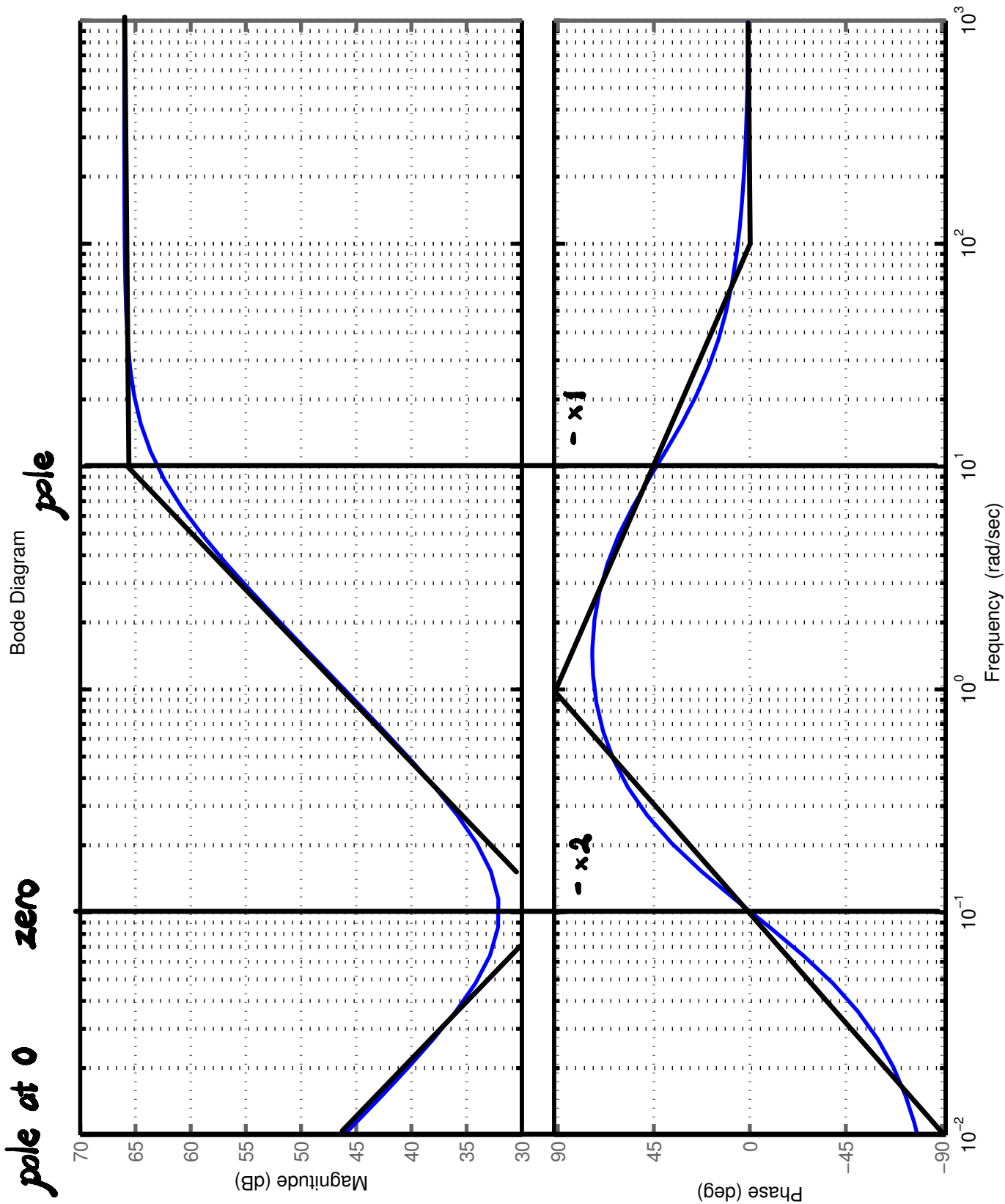


Figure 3.B

4. Consider a system that delays its input by time T , i.e. a system whose input-output behaviour is $y(t) = x(t - T)$. In Tutorial Sheet 3 we found that the transfer function for this system is

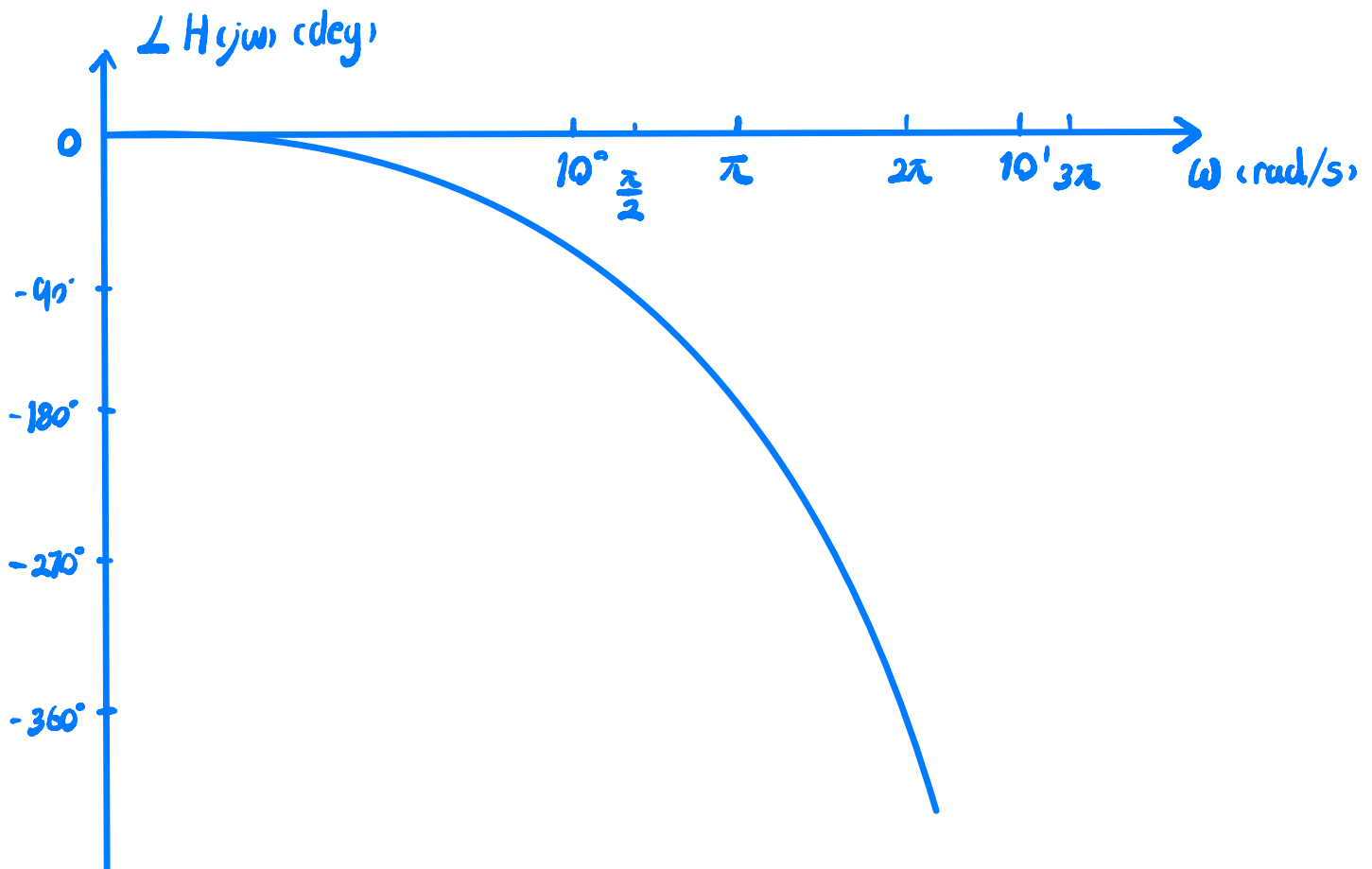
$$H(s) = e^{-sT}.$$

Draw the Bode diagram for this system.

$$H(j\omega) = e^{-j\omega T} = \cos(\omega T) - j\sin(\omega T)$$

$$|H(j\omega)| = 1 = 0 \text{ dB for any frequencies}$$

for $\angle H(j\omega)$.



5. Suppose that the signal $y(t)$ is generated via convolution of two signals

$$y(t) = x_1(t) * x_2(t),$$

where

$$X_1(j\omega) = 0, \quad \text{for } |\omega| > 1000\pi \quad (1)$$

$$X_2(j\omega) = 0, \quad \text{for } |\omega| > 2000\pi. \quad (2)$$

If impulse train sampling is used to generate the signal

$$y_p(t) = \sum_{n=-\infty}^{+\infty} y(nT)\delta(t - nT),$$

specify the range of values T which ensure that $y(t)$ is recoverable from $y_p(t)$.

$$y_p(t) = \sum_{n=-\infty}^{+\infty} y(nT)\delta(t - nT)$$

$$= y(t)$$

$$= x_1(t) * x_2(t)$$

$$\therefore Y(j\omega) = X_1(j\omega) \cdot X_2(j\omega)$$

$$= \begin{cases} 0 & |\omega| > 1000\pi \\ X_1(j\omega) \cdot X_2(j\omega) & |\omega| < 1000\pi \end{cases}$$

see solutions, $\omega > 2000\pi$